Hangxu (Levi) Sha

Richmond, BC | 236-513-1888 | hangxu.sha9@gmail.com | linkedin.com/in/levi-sha-505679382

SUMMARY OF QUALIFICATIONS

- **Electrical & Computing Engineering Technology** student at BCIT with a strong foundation in circuit design, schematic capture, and PCB fabrication.
- **Hands-on experience** developing hardware projects, utilizing KiCad, LTspice, SolidWorks, and CPLD programming.
- **Proficient in prototyping,** precision soldering, PCB assembly, and hardware debugging using Digital Multimeters (DMM).
- **Strong team collaborator** and analytical thinker with a proven track record in community volunteering and project delivery.

TECHNICAL SKILLS

- **CAD & Simulation Software:** KiCad, LTspice, SolidWorks, AutoCAD, SketchUp, Falstad
- **Hardware & Fabrication:** PCB Design & Assembly, Circuit Design, Through-Hole & SMD Soldering, Hardware Troubleshooting, DMM Debugging
- **Programming Languages:** Python, C/C++, SystemVerilog, Java, AHDL, HTML/CSS (Learning)
- **Tools & Productivity:** Excel (Data Analysis & Formula Modeling)

EDUCATION

Diploma in Electrical and Computing Engineering Technology — **Sept 2025 – Present (Expected May 2029)**
British Columbia Institute of Technology (BCIT)

Dogwood Diploma — **Sept 2021 – Jan 2025**
Canada Star Secondary School

TECHNICAL PROJECTS

AC to DC Power Supply — **Sept 2025 – Present**
BCIT ELEX Fabrication Course

- Designed a complete schematic and multi-layer PCB layout using KiCad.
- Populated components via precision soldering and systematically debugged hardware connections using a DMM.
- Utilized SolidWorks to modify and optimize the physical chassis enclosure for component fitment.
- Engineered the unit to safely convert 120V AC to variable/constant DC, integrating a built-in fuse and protective grounding protocols.

Logic Gate RC Car
Jan 2026 – Present

BCIT Electronic Circuits Course

- Modeled and simulated complex circuit behaviors within LTspice to validate design parameters before physical prototyping.
- Implemented an H-bridge motor driver circuit to achieve bi-directional wheel control.
- Utilized MOSFETs as high-speed switches to dynamically adjust operating frequencies and optimize power efficiency.

Java Platformer Game & Engine
Sept 2023 – Jan 2024

Canada Star Secondary School Capstone Project

- Built a custom 2D game engine from scratch utilizing Object-Oriented Programming (OOP) in Java.
- Programmed core mechanics including physics simulation (gravity, collision detection) and user input handling.
- Optimized runtime update loops to resolve rendering bottlenecks and ensure smooth gameplay execution.

VOLUNTEER EXPERIENCE

Volunteer Coordinator
Apr 2022 – May 2023

Social Diversity for Children Foundation

- Dedicated 25+ hours per month to coordinate fundraising operations, managing logistics and public-facing sales at transit hubs.
- Co-hosted charity events accommodating dozens of community attendees, improving communication and organizational agility.

Camp Assistant
Feb 2022

Spring Science Camp

- Mentored young students on foundational engineering concepts, guiding the structural design of mock Mars rovers.
- Fostered a collaborative team environment to help students overcome design challenges during physics-based projects.